User Creation and Privilege Management

Module Title and Code: Introduction to Databases, UFCFTK-30-1

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TASK 3 (20%)

Creating Users and Granting Privileges (Based on Task 2 Tables)

Write SQL commands to:

1. Create a user data_analyst with the password Analyst@123.

Answer:

CREATE USER 'data_analyst'@'%' IDENTIFIED BY 'Analyst@123';

```
Setting environment for using XAMPP for Windows.
rauna@BATMAN c:\xampp
# mysql -u root -p
Enter password: ***********
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 11
Server version: 10.4.32-MariaDB mariadb.org binary distribution

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE tracker;
Database changed
MariaDB [tracker]> CREATE USER 'data_analyst'@'%' IDENTIFIED BY 'Analyst@123';
Query OK, 0 rows affected (0.009 sec)
```

A new user named **data_analyst** was created with a password.
This allows controlled access to the database system.

2. Grant the user data_analyst SELECT privileges on all tables in the TRACKER database.

Answer:

GRANT SELECT ON TRACKER.* TO 'data_analyst'@'%';

```
MariaDB [tracker]> GRANT SELECT ON TRACKER.* TO 'data_analyst'@'%';
Query OK, 0 rows affected (0.005 sec)
```

The **SELECT** permission was given to the user on all tables in the **TRACKER** database. This means the user can view the data but cannot make any changes.

3. Grant the user data_analyst INSERT privileges only on the FLIGHT table in the TRACKER database.

Answer:

GRANT INSERT ON TRACKER.FLIGHT TO 'data_analyst'@'%';

```
MariaDB [tracker]> GRANT INSERT ON TRACKER.FLIGHT TO 'data_analyst'@'%';
Query OK, 0 rows affected (0.004 sec)
```

The **INSERT** permission was given only on the **FLIGHT** table. This allows the user to add new records, but only in that specific table.

4. Revoke the INSERT privilege from the FLIGHT table.

Answer:

REVOKE INSERT ON TRACKER.FLIGHT FROM 'data_analyst'@'%';

```
MariaDB [tracker]> REVOKE INSERT ON TRACKER.FLIGHT FROM 'data_analyst'@'%';
Query OK, 0 rows affected (0.004 sec)
```

The **INSERT** permission was later removed from the **FLIGHT** table using the **REVOKE** command. This shows how access can be changed when needed.

Proof of Execution:

The commands were verified by checking the user details and privileges using SQL queries like SHOW GRANTS. This confirms that the user was created and permissions were applied correctly.

1. Checking if user exists:

SELECT User, Host FROM mysql.user WHERE User = 'data_analyst';

```
MariaDB [tracker]> SELECT User, Host FROM mysql.user WHERE User = 'data_analyst';
+--------------+------+
| User         | Host |
+--------------+------+
| data_analyst | %    |
+--------------+------+
1 row in set (0.008 sec)
```

2. Checking granted privileges:

SHOW GRANTS FOR 'data_analyst'@'%';

```
MariaDB [tracker]> SHOW GRANTS FOR 'data_analyst'@'%';
+-----------------------------------------------------------------------------------------------------------------+
| Grants for data_analyst@%                                                                                       |
+-----------------------------------------------------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `data_analyst`@`%` IDENTIFIED BY PASSWORD '*F7B87AEFE42254FF826AC55E6E9B92A04F7CD3CB' |
| GRANT SELECT ON `tracker`.* TO `data_analyst`@`%`                                                               |
+-----------------------------------------------------------------------------------------------------------------+
2 rows in set (0.001 sec)
```